

A PARTIAL RESOLUTION OF ERDŐS PROBLEM 124

AUTONOMOUS AI ASSISTANT AND HUMAN COLLABORATOR

ABSTRACT. Erdős Problem 124 asks whether for any finite set $A = \{d_1, \dots, d_r\} \subseteq \mathbb{Z}_{\geq 2}$ with $\gcd(A) = 1$ and $\sum \frac{1}{d_i-1} \geq 1$, the subset sums of the powers d_i^e for $e \geq k$ represent all sufficiently large integers. We present a partial resolution of this problem. First, we prove the base-2 case affirmatively by modular residue absorption. Second, we reformulate the $k \geq 1$ problem for arbitrary bases as a finite deletion from the $k = 0$ Brown-complete channelized sequence. Under this framework, we prove a strict-surplus finite-deletion bootstrap theorem for sets with $\sum \frac{1}{d_i-1} > 1$: if a finite prefix establishes a central contiguous interval of radius C , all later powers are automatically absorbed past an explicitly computable threshold T_C . Using this theorem, we provide exact finite certificates of cofiniteness for the families containing $\{3, 4, 5\}$ and $\{3, 4, 6\}$ at $k = 1$. Finally, we isolate the remaining exact-critical obstruction to the Central Seed Problem, demonstrating that failure to absorb forces a bounded additive near-collision.

1. INTRODUCTION

For a set $S \subseteq \mathbb{Z}_{\geq 1}$, let $\Sigma(S)$ denote the set of subset sums of S :

$$\Sigma(S) = \left\{ \sum_{s \in I} s : I \subseteq S, |I| < \infty \right\}.$$

A set S is called *cofinite* if it represents all sufficiently large integers (i.e., $\mathbb{Z}_{\geq c} \subseteq \Sigma(S)$ for some conductor c).

Erdős Problem 124 concerns the subset sums of perfect powers. For a finite set of bases $A = \{d_1, \dots, d_r\}$, the official problem is defined over the multiset of powers generated by each base independently. To formalize this without ambiguity regarding overlapping powers (e.g., $3^4 = 9^2$), we define the **channelized multiset** starting from exponent k :

$$\mathcal{P}_A(k) = \{(i, e) : 1 \leq i \leq r, e \geq k\}$$

with weight $w(i, e) = d_i^e$. A representation is $N = \sum_{(i,e) \in I} d_i^e$ where $I \subset \mathcal{P}_A(k)$ is finite.

The standard density heuristic is given by $R(A) = \sum_{i=1}^r \frac{1}{d_i-1}$. If $\gcd(A) = 1$ and $R(A) \geq 1$, the sequence is conjectured to be cofinite.

2. THE BASE-2 CASE

When $2 \in A$, the problem can be resolved completely via elementary modular arithmetic, independent of the density condition.

Proposition 2.1. *If $2 \in A$ and $\gcd(A) = 1$, then for every $k \geq 1$, $\mathcal{P}_A(k)$ is cofinite.*

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Proof. Since $\gcd(A) = 1$ and $2 \in A$, the set A must contain at least one odd base b . By Euler's Totient Theorem, the sequence of powers modulo 2^k is periodic, and $b^{\phi(2^k)} \equiv 1 \pmod{2^k}$. There are infinitely many exponents $e \geq k$ such that $b^e \equiv 1 \pmod{2^k}$.

Select exactly $2^k - 1$ distinct channelized powers of b , each congruent to 1 $\pmod{2^k}$. Let this finite subset be S_{odd} . Let $M_{\text{odd}} = \Sigma(S_{\text{odd}})$ be its maximum subset sum.

By construction, taking subset sums of S_{odd} generates exactly the values $0, 1, 2, \dots, 2^k - 1$ modulo 2^k . Therefore, for any integer $N > M_{\text{odd}}$, there exists a specific subset sum $S \subseteq S_{\text{odd}}$ such that $S \equiv N \pmod{2^k}$.

Consider the difference $N - S$. Because $N > M_{\text{odd}} \geq S$, the difference is strictly positive and is a multiple of 2^k : $N - S = m \cdot 2^k$ for some positive integer m . Every positive integer m possesses a unique binary representation. Therefore, $m \cdot 2^k$ can be uniquely expressed as a subset sum of the powers of 2 starting from exponent k . Since the channels are disjoint, we can combine these subset sums to form N . Thus, the sequence is unconditionally cofinite. $\square$

3. FINITE-DELETION FROM THE $k = 0$ SEQUENCE

For $k = 0$, every base contributes $1 = d_i^0$. If we sort all channelized powers as $a_1 \leq a_2 \leq \dots$, Brown's criterion states that if $a_{n+1} \leq 1 + \sum_{j \leq n} a_j$ for all n , then all positive integers are represented. For $k = 0$, if $R(A) \geq 1$, the mass generated strictly satisfies this inequality, making $\mathcal{P}_A(0)$ unconditionally cofinite [1].

For $k \geq 1$, the removed powers are exactly the finite initial blocks $d_i^0, d_i^1, \dots, d_i^{k-1}$. The problem is thereby reframed: does deleting a finite initial block from each channel preserve cofiniteness?

3.1. Strict-Surplus Finite-Deletion Bootstrap. Let $C_0(A, k) = \sum_{i=1}^r \frac{d_i^k}{d_i - 1}$. At any stage in the $k \geq 1$ sequence, let E_i be the next unused power in channel i , and let $t = \min_i E_i$. The previous channel mass is exactly $\sigma = \sum_{i=1}^r \frac{E_i - d_i^k}{d_i - 1}$.

Therefore:

$$\begin{aligned} \sigma - t &= \sum_i \frac{E_i - t}{d_i - 1} + t \sum_i \frac{1}{d_i - 1} - C_0(A, k) - t \\ \sigma - t &= (R(A) - 1)t - C_0(A, k) + \sum_i \frac{E_i - t}{d_i - 1} \end{aligned}$$

Since every $E_i \geq t$, we obtain the fundamental mass estimate:

$$\sigma - t \geq (R(A) - 1)t - C_0(A, k)$$

Define a finite prefix F to have a *central radius* C if $[C, \sigma(F) - C] \subseteq \Sigma(F)$.

Theorem 3.1 (Strict-Surplus Bootstrap). *Assume $R(A) > 1$. Suppose some finite prefix $F \subset \mathcal{P}_A(k)$ has central radius C . If all next powers up to*

$$T_C = \frac{2C - 1 + C_0(A, k)}{R(A) - 1}$$

have been checked and successfully absorbed, then every later power is automatically absorbed, and $\mathcal{P}_A(k)$ is cofinite.

Proof. If the current central interval is $[C, \sigma - C]$, then after appending the next power t , the shifted interval is $[C + t, \sigma - C + t]$. The two intervals touch or overlap to form a new contiguous central interval of radius C if and only if:

$$C + t \leq \sigma - C + 1 \implies t \leq \sigma - 2C + 1$$

Using the mass estimate, $\sigma - t \geq (R(A) - 1)t - C_0(A, k)$. Thus, $t \leq \sigma - 2C + 1$ is algebraically guaranteed once $\sigma - t \geq 2C - 1$, which requires:

$$(R(A) - 1)t - C_0(A, k) \geq 2C - 1 \implies t \geq \frac{2C - 1 + C_0(A, k)}{R(A) - 1} = T_C$$

After this finite threshold T_C , the central radius C persists forever, and the contiguous interval grows without bound. $\square$

3.2. Exact Finite Certificates. Using exact subset-sum bitset computations, Theorem 3.1 yields immediate rigorous proofs for specific families.

Corollary 3.2. *The core $\{3, 4, 5\}$ at $k = 1$ is cofinite.*

Proof. Here $R = \frac{13}{12} > 1$, and $C_0 = \frac{49}{12}$. A finite prefix through 1024 yields $\sigma = 3236$ and central radius $C = 80$. The threshold is

$$T_C = \frac{2(80) - 1 + 49/12}{1/12} = 1957.$$

The next power after the prefix is $3^7 = 2187$. Since $2187 > 1957$, the infinite tail is automatic. $\square$

Corollary 3.3. *The core $\{3, 4, 6\}$ at $k = 1$ is cofinite.*

Proof. Here $R = \frac{31}{30} > 1$, and $C_0 = \frac{121}{30}$. A finite prefix through 59049 yields $\sigma = 166402$ and central radius $C = 987$. The threshold is

$$T_C = \frac{2(987) - 1 + 121/30}{1/30} = 59311.$$

The next power is $4^8 = 65536 > 59311$. The tail is automatic. $\square$

Any larger base set A containing one of these certified cores is also cofinite, as extra available powers cannot destroy existing representations.

3.3. The Critical Case ($R(A) = 1$). For exact-critical sets, the threshold T_C diverges. However, if central absorption fails ($t > \sigma - 2C + 1$), then $\sigma - t < 2C - 1$. Substituting the mass equality $\sigma - t = \sum_i \frac{E_i - t}{d_i - 1} - C_0(A, k)$, we derive:

$$\sum_i \frac{E_i - t}{d_i - 1} < 2C - 1 + C_0(A, k)$$

Since all terms are non-negative, for any base i :

$$0 \leq E_i - t < (d_i - 1)(2C - 1 + C_0(A, k))$$

Thus, in the critical case, *after a central seed of radius C is established*, every future absorption failure genuinely forces bounded additive near-collisions among the next channel powers.

4. CONCLUSION

We resolve the base-2 case of Erdős Problem 124 by modular residue absorption. We reformulate the $k \geq 1$ problem as finite deletion from the $k = 0$ Brown-complete sequence, proving a strict-surplus finite-deletion bootstrap theorem. This yields exact finite certificates for strict-surplus cores such as $\{3, 4, 5\}$ and $\{3, 4, 6\}$.

We isolate the remaining exact-critical obstruction as the **Central Seed Problem**. If a central interval can be established, the Diophantine tail is restricted to finite S -unit near-collisions [2]. This cleanly separates the combinatorial seed problem from the analytic Diophantine tail.

REFERENCES

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